

Customer Shopping Behavior Analysis

Easy-to-Understand Project Report

Project focus: Understand customer spending, product choices, discounts, subscriptions and loyalty.

Area	Used in the project
Python	Pandas, data cleaning, missing-value handling and feature creation
SQL	Business questions, grouping, ranking, filtering and calculations
Power BI	Interactive dashboard and visual reporting
Dataset	3,900 customer purchase records and 18 columns

This report is based on the provided Python notebook, SQL query file, project report and Power BI dashboard. The wording is simplified for easy understanding in an interview or viva.

1. Project Overview

This project studies customer shopping data to understand how customers spend money, what products they buy, how discounts are used, subscription behavior and customer loyalty.

In simple words: **Python was used to clean the data, SQL was used to answer business questions, and Power BI was used to show the results visually.**

2. Dataset in Simple Words

Item	Details
Total records	3,900
Total columns	18
Customer information	Age, Gender, Location and Subscription Status
Purchase information	Item Purchased, Category, Purchase Amount, Season, Size and Color
Shopping behavior	Discount Applied, Previous Purchases, Purchase Frequency, Review Rating and Shipping Type
Missing values	37 missing values were found in Review Rating

Missing Review Rating values were handled using the median rating of the relevant product category.

3. Python Data Cleaning and Preparation

The Python notebook uses Pandas to prepare the data before SQL analysis.

- **Load data:** Read the customer CSV using pandas.
- **Explore data:** Used head(), info() and describe() to understand the dataset.
- **Check missing values:** Checked null values in all columns.
- **Fill missing ratings:** Used the median rating of each category.
- **Clean column names:** Converted names to lowercase and underscores.
- **Create age groups:** Divided customers into useful age groups.
- **Create purchase frequency:** Converted purchase frequency text into days.
- **Remove duplicate information:** Checked discount and promo-code fields and dropped promo_code_used.
- **Database connection:** Loaded the cleaned data into the database for SQL analysis.

4. SQL Analysis – Business Questions

The SQL file contains 10 main questions. Each question answers a practical business problem.

Q	Business question	What it tells us
1	Revenue by gender?	Which gender generated more revenue.
2	Who used discounts but spent above average?	Finds high-spending discount users.
3	Top 5 products by rating?	Finds highly rated products.
4	Standard vs Express average purchase?	Compares average spending by shipping type.
5	Do subscribers spend more?	Compares average spend and total revenue.
6	Which products have the most discounts?	Finds discount-heavy products.
7	How many New, Returning and Loyal customers?	Segments customers using purchase history.
8	Top 3 products in each category?	Finds the most purchased products by category.
9	Are repeat buyers more likely to subscribe?	Checks subscription among repeat buyers.
10	Revenue by age group?	Shows revenue contribution by age group.

SQL concepts used: GROUP BY, SUM(), AVG(), COUNT(), CASE, subqueries, ORDER BY, LIMIT, conditional aggregation and DENSE_RANK().

5. Important SQL Results

Analysis	Result	Simple meaning
Revenue by gender	Male: \$157,890 Female: \$75,191	Male customers generated more total revenue in this dataset.
Discount + high spending	839 records	Customers used discounts and still spent above the overall average.
Top rated products	Gloves 3.86 Sandals 3.84 Boots 3.82 Hat 3.80 Skirt 3.78	These five products had the highest average ratings.
Shipping	Standard: \$58.46 Express: \$60.48	Express customers had a slightly higher average purchase.
Subscribers	1,053 customers Avg: \$59.49 Revenue: \$62,645	Subscribers generated less total revenue because there were fewer of them.
Non-subscribers	2,847 customers Avg: \$59.87 Revenue: \$170,436	Non-subscribers generated more total revenue because they were more numerous.
Discount-heavy products	Hat 50.00% Sneakers 49.66% Coat 49.07% Sweater 48.17% Pants 47.37%	These products had the highest percentage of discounted purchases.
Customer segments	Loyal: 3,116 Returning: 701 New: 83	Most customers are in the Loyal segment based on previous purchases.
Repeat buyers >5	No: 2,518 Yes: 958	More repeat buyers were non-subscribers than subscribers.
Age-group revenue	Young Adult: \$62,143	Young Adult was the highest revenue group in the project result.

6. Power BI Dashboard

The dashboard presents the analysis in a visual and interactive way. Users can filter the results by subscription status, gender, shipping type and category.



Provided Power BI dashboard screenshot.

Dashboard item	Easy explanation
3.9K Customers	Total customer records shown in the dashboard.
\$59.76 Average Purchase	Average amount spent in one purchase.
3.75 Average Rating	Average product review rating.

Dashboard item	Easy explanation
Subscription Status	About 27% are subscribed and 73% are not subscribed in the dashboard.
Revenue by Category	Shows revenue for Clothing, Accessories, Footwear and Outerwear.
Sales by Category	Shows sales comparison between product categories.
Filters	Allows the user to view selected customer or purchase groups.

7. Project Flow – Easy Explanation

Step	What I did	Why I did it
1. Load	Loaded customer data in Python.	To bring raw data into the analysis.
2. Check	Used info(), describe() and null checks.	To understand the data and find issues.
3. Clean	Filled missing ratings and standardized names.	To make the data cleaner and reliable.
4. Transform	Created age groups and purchase frequency in days.	To make customer behavior easier to analyze.
5. Store	Connected Python with the database.	To perform SQL analysis.
6. Analyze	Answered 10 business questions with SQL.	To turn raw data into useful answers.
7. Visualize	Built the Power BI dashboard.	To show important results clearly.
8. Recommend	Suggested business actions.	To connect data findings with decisions.

8. Business Recommendations

- **Increase subscriptions:** Give useful benefits or exclusive offers to encourage more customers to subscribe.
- **Reward loyal customers:** Provide loyalty benefits to repeat buyers.
- **Control discount dependency:** Review products with very high discount usage so discounts do not reduce profit unnecessarily.
- **Promote strong products:** Highlight highly rated and best-selling products.
- **Target valuable groups:** Focus marketing on customer groups and categories that generate higher revenue.

9. Conclusion

This project demonstrates a complete analytics workflow: **Python** → **SQL** → **Power BI** → **Business Recommendations**.

Python prepares the data, SQL answers business questions, and Power BI makes the results easy to understand. The analysis shows a large Loyal customer group, strong performance from Clothing, high discount usage for some products, and a much larger non-subscriber customer base.

The project report recommends improving subscriptions, rewarding repeat customers, reviewing discount policies, promoting strong products and using targeted marketing.

Note: The Python notebook and SQL query use different age-group methods. Python creates four groups using qcut(), while SQL uses fixed age ranges. Both are kept as provided in the original project.